CHAPTER 15

TRANSPORT AND COMMUNICATION IN INDIA

CHAPTER ESSENCE AT A GLANCE:
Master India transport and telecommunications: the Golden Quadrilateral and Corridor networks, Border Roads Organisation (BRO), railway track gauges and zones, Konkan Railway engineering marvel, National Waterways (NW-1, NW-2, NW-3 in Kerala), pipelines (HVJ), and INSAT and IRS satellites.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Road Networks and High-Speed Corridors:** India has the second largest road network in the world (about 63 lakh km). The Golden Quadrilateral (5,846 km) connects Delhi, Mumbai, Chennai, and Kolkata. North-South Corridor (Srinagar to Kanyakumari) and East-West Corridor (Silchar to Porbandar) intersect at Jhansi.

• **Border Roads Organisation (BRO):** Established in May 1960 to construct and maintain strategic roads along the northern and north-eastern frontiers, facilitating national defence and economic development; constructed Atal Tunnel and high-altitude Himalayan passes.

• **Indian Railways and Gauges:** Introduced in 1853 between Mumbai and Thane (34 km). Three track gauges: Broad Gauge (1.676 m, 90+ percent of route), Metre Gauge (1.000 m), and Narrow Gauge (0.762 m and 0.610 m). Konkan Railway (760 km, Roha to Thokur) traverses 146 rivers and 91 tunnels across the Western Ghats.

• **Inland Waterways, Pipelines and Satellites:** Inland navigation: NW-1 (Ganga: Prayagraj to Haldia, 1,620 km), NW-2 (Brahmaputra: Sadiya to Dhubri, 891 km), and NW-3 (West Coast Canal in Kerala: Kottapuram to Kollam, 205 km). Hazira-Vijaipur-Jagdishpur (HVJ) gas pipeline. Space systems: INSAT (telecommunication) and IRS (remote sensing).

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

$$\text{Golden Quadrilateral Super Highway} \Longleftrightarrow \text{Delhi} - \text{Mumbai} - \text{Chennai} - \text{Kolkata} \quad [5,846\ \text{km}]$$
Four Metropolitan Super Highway Quadrilateral

$$\text{National Waterway-1 (NW-1)} \Longleftrightarrow \text{Ganga-Bhagirathi-Hooghly Route} \quad [\text{Prayagraj to Haldia, } 1{,}620\ \text{km}]$$
Premier Navigable Inland Water Artery

$$\text{National Waterway-3 (NW-3)} \Longleftrightarrow \text{West Coast Canal} \quad [\text{Kottapuram to Kollam, } 205\ \text{km}]$$
Kerala Navigable Backwater Inland Artery

$$\text{Indian Space Telecommunication} \Longleftrightarrow \text{INSAT System (Telecommunications)} + \text{IRS System (Remote Sensing)}$$
Dual Satellite Space Communication Architecture

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** The North-South and East-West National Highway corridors intersect at Jhansi in Uttar Pradesh.

• **Alert:** Konkan Railway passes through Maharashtra, Goa, and Karnataka, providing direct economic connectivity from Roha to Thokur (Mangalore) and benefiting Kerala.

• **Alert:** National Waterway-3 (NW-3) is situated entirely within Kerala along the West Coast Canal from Kottapuram to Kollam, including Champakara and Udyogmandal canals.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Describe the Golden Quadrilateral Super Highway, mentioning its terminals, total length, and economic objectives. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Terminals:** A 4 to 6 lane high-speed highway linking India four premier metropolitan cities: Delhi, Mumbai, Chennai, and Kolkata. **[1 Mark]**

• **Total Length:** Spans a total length of 5,846 km, administered by the National Highways Authority of India (NHAI). **[1 Mark]**

• **Economic Objectives:** Drastically curtails transit time and freight transportation costs between major industrial and urban centres of India. **[1 Mark]**

2. Explain the location and significance of the North-South Corridor and East-West Corridor, and name their junction point. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **North-South Corridor:** Spans 4,076 km connecting Srinagar in Jammu and Kashmir to Kanyakumari in Tamil Nadu. **[1 Mark]**

• **East-West Corridor:** Spans 3,640 km connecting Silchar in Assam to the port town of Porbandar in Gujarat. **[1 Mark]**

• **Intersection Point:** The two national super-highway corridors intersect at Jhansi in Uttar Pradesh. **[1 Mark]**

3. Highlight the vital contributions of the Border Roads Organisation (BRO) in India border areas. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Establishment:** Established in May 1960 for the rapid, coordinated development of roads in northern and northeastern border states. **[1 Mark]**

• **Defence Preparedness:** Ensures year-round strategic movement of military defence personnel and heavy supplies to high-altitude frontier outposts. **[1 Mark]**

• **Engineering Feats:** Constructed world-record high-altitude motorable roads across Leh-Manali and built the engineering landmark Atal Tunnel under Rohtang Pass. **[1 Mark]**

4. Explain the significance of the Konkan Railway in connecting the western coastal plain of India, highlighting its engineering challenges. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Route and Extent:** Extends 760 km from Roha in Maharashtra to Thokur near Mangalore in Karnataka, passing through Goa. **[1.5 Marks]**

• **Engineering Feats:** Overcame formidable Western Ghats obstacles, constructing 146 major river bridges, 2,000 minor bridges, and 91 mountain tunnels (including the 6.5 km Karbude tunnel). **[1.5 Marks]**

• **Economic Impact:** Drastically reduced travel time between Mumbai and Kerala from over 40 hours to under 24 hours, stimulating coastal trade and tourism. **[1 Mark]**

5. Describe the three track gauges used in Indian Railways with their precise widths. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **1. Broad Gauge:** Distance between rails is 1.676 metres; accounts for over 90 percent of the total Indian railway route network under Project Unigauge. **[1 Mark]**

• **2. Metre Gauge:** Distance between rails is 1.000 metre; progressively converted into broad gauge across the country. **[1 Mark]**

• **3. Narrow Gauge:** Distance between rails is 0.762 metre or 0.610 metre; preserved primarily in scenic mountain railways like Darjeeling and Nilgiri. **[1 Mark]**

6. Provide the route details, terminals, and length of National Waterway-1 (NW-1) and National Waterway-3 (NW-3). **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **National Waterway-1 (NW-1):** Navigable along the Ganga-Bhagirathi-Hooghly river system between Prayagraj and Haldia, covering 1,620 km. **[1.5 Marks]**

• **National Waterway-3 (NW-3):** Located in Kerala along the West Coast Canal, extending 205 km between Kottapuram and Kollam (including Champakara and Udyogmandal canals). **[1.5 Marks]**

7. Distinguish between the INSAT satellite system and the IRS satellite system of India. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **INSAT (Indian National Satellite System):** Geostationary telecommunication satellites used for telephone connectivity, television broadcasting, and real-time cyclone and meteorological disaster warning. **[1.5 Marks]**

• **IRS (Indian Remote Sensing Satellite System):** Sun-synchronous satellites carrying multi-spectral sensors for mapping land use, forest cover, groundwater reserves, and mineral exploration. **[1.5 Marks]**